

Summary

- Every idempotent matrix $\mathbf{P}^2 = \mathbf{P}$ represents a projection. If, in addition, the matrix is self-adjoint $\langle \mathbf{P}\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{P}\mathbf{y} \rangle$, then the projection is orthogonal.
- If V is expressible as the direct sum of two subspaces $V_1 \oplus V_2$, then the projection onto V is the sum of the projections onto V_1 and onto V_2 .
- The orthogonal projection $\hat{\mathbf{u}}$ of $\mathbf{u}$ onto a subspace V has two defining properties: (i.) $\hat{\mathbf{u}} \in V$ and (ii.) $\mathbf{u} - \hat{\mathbf{u}} \in V^\perp$.
- Every finite dimensional subspace V of a Hilbert space $\mathcal{H}$ is closed.
- If V is a closed subspace of $\mathcal{H}$, then for any $\mathbf{u} \in \mathcal{H}$, there exists a unique closest element in V to $\mathbf{u}$, and that element is the orthogonal projection $\hat{\mathbf{u}}$.

Direct Sums

Definition 2.1. Suppose V_1 and V_2 are linear subspaces. The sum $V = V_1 + V_2$ consists of those vectors $\{\mathbf{v}_1 + \mathbf{v}_2 : \mathbf{v}_1 \in V_1, \mathbf{v}_2 \in V_2\}$. If V_1 and V_2 are linearly independent, then V is described as the **direct sum** of V_1 and V_2 , written $V = V_1 \oplus V_2$. ■

Proposition 2.1. If $V = V_1 \oplus V_2$, then any $\mathbf{v} \in V$ has a unique representation $\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2$ where $\mathbf{v}_1 \in V_1$ and $\mathbf{v}_2 \in V_2$. ♦

Proof. Since V is the sum of V_1 and V_2 , there exist $\mathbf{v}_1 \in V_1$ and $\mathbf{v}_2 \in V_2$ such that $\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2$. Suppose that there exists another representation $\mathbf{v} = \mathbf{u}_1 + \mathbf{u}_2$, then:

$$\mathbf{0} = \mathbf{v} - \mathbf{v} = (\mathbf{v}_1 - \mathbf{u}_1) + (\mathbf{v}_2 - \mathbf{u}_2).$$

But $\mathbf{v}_1 - \mathbf{u}_1 \in V_1$ and $\mathbf{v}_2 - \mathbf{u}_2 \in V_2$. By linear independence of V_1 and V_2 , $\mathbf{v}_1 - \mathbf{u}_1 = \mathbf{0}$ and $\mathbf{v}_2 - \mathbf{u}_2 = \mathbf{0}$. Conclude the representation of $\mathbf{v}$ is unique. ■

Definition 2.2. A **basis** is a collection B of *linearly independent* vectors that *span* a linear space V . ■

Proposition 2.2. Suppose B_1 is a basis for V_1 and B_2 is a basis for V_2 , then the union $B = B_1 \cup B_2$ is a basis for $V = V_1 \oplus V_2$. ♦

Proof. Any $\mathbf{v} \in V$ is uniquely expressible as $\mathbf{v}_1 + \mathbf{v}_2$, where $\mathbf{v}_1 \in V_1$ and $\mathbf{v}_2 \in V_2$. Since B_1 and B_2 are bases for V_1 and V_2 , $\mathbf{v}_1$ has a unique representation with respect to B_1 , and $\mathbf{v}_2$ has a unique representation with respect to B_2 . Thus $\mathbf{v}$ is in the span of $B_1 \cup B_2$. Moreover, since V_1 and V_2 are linearly independent, and B_1 and B_2 are each bases, the collection of vectors $B_1 \cup B_2$ is linearly independent. ■

Corollary 2.1. If $V = V_1 \oplus V_2$, then $\dim(V) = \dim(V_1) + \dim(V_2)$. ♣

Projections

Definition 3.1. Suppose $V = V_1 \oplus V_2$, then every $\mathbf{v} \in V$ has a uniform representation $\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2$ where $\mathbf{v}_1 \in V_1$ and $\mathbf{v}_2 \in V_2$. The mapping $\Pi(\mathbf{v}|V_1) = \mathbf{v}_1$ is the **projection** of $\mathbf{v}$ onto V_1 . Likewise, $\Pi(\mathbf{v}|V_2) = \mathbf{v}_2$ is the projection of $\mathbf{v}$ onto V_2 . ■

Proposition 3.1. Suppose $V = V_1 \oplus V_2$, and $\mathbf{v} \in V$. The projection $\Pi(\mathbf{v}|V_1)$ of $\mathbf{v}$ onto V_1 is a linear mapping. ♦

Proof. Suppose $\mathbf{u}$ and $\mathbf{v}$ are in V , then $\mathbf{u} = \mathbf{u}_1 + \mathbf{u}_2$ and $\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2$, with $\mathbf{u}_k, \mathbf{v}_k \in V_k$. Consider the sum $\mathbf{w} \equiv \mathbf{u} + \mathbf{v} = (\mathbf{u}_1 + \mathbf{v}_1) + (\mathbf{u}_2 + \mathbf{v}_2)$. Observe that $\mathbf{w}_k = \mathbf{u}_k + \mathbf{v}_k \in V_k$. Since this representation is unique:

$$\Pi(\mathbf{u} + \mathbf{v}|V_1) = \Pi(\mathbf{w}|V_1) = \mathbf{w}_1 = \mathbf{u}_1 + \mathbf{v}_1 = \Pi(\mathbf{u}|V_1) + \Pi(\mathbf{v}|V_1).$$

■

Corollary 3.1. $\Pi(\cdot|V_1)$ is expressible as a *projection matrix* $\mathbf{P}$. ♣

Proposition 3.2. Suppose $V = V_1 \oplus V_2$, and $\mathbf{v} \in V$. The projection $\Pi(\mathbf{v}|V_1)$ mapping is **idempotent**. Consequently, $\mathbf{P}^2 = \mathbf{P}$. ♦

Proof. The projection of $\mathbf{v}$ onto V_1 is:

$$\Pi(\mathbf{v}|V_1) = \mathbf{v}_1 = \mathbf{P}\mathbf{v}.$$

Since $\mathbf{v}_1$ already belongs to V_1 , upon projecting again:

$$\mathbf{P}\mathbf{P}\mathbf{v} = \mathbf{P}\mathbf{v}_1 = \Pi(\mathbf{v}_1|V_1) = \mathbf{v}_1 = \mathbf{P}\mathbf{v}.$$

Since $\mathbf{P}^2\mathbf{v} = \mathbf{P}\mathbf{v}$ holds for $\forall \mathbf{v} \in V$, conclude that $\mathbf{P}^2 = \mathbf{P}$. ■

Proposition 3.3. Suppose $V = V_1 \oplus V_2$, and $\mathbf{v} \in V$. Let $\mathbf{P}$ denote projection onto V_1 . Then, $\mathbf{Q} = \mathbf{I} - \mathbf{P}$ is projection onto V_2 and $\mathbf{P}\mathbf{Q} = \mathbf{Q}\mathbf{P} = \mathbf{0}$. ♦

Proof. The projection onto V_2 is:

$$\Pi(\mathbf{v}|V_2) = \mathbf{v}_2 = \mathbf{v} - \mathbf{v}_1 = \mathbf{v} - \Pi(\mathbf{v}|V_1) = \mathbf{v} - \mathbf{P}\mathbf{v} = (\mathbf{I} - \mathbf{P})\mathbf{v}.$$

Let $\mathbf{Q} = \mathbf{I} - \mathbf{P}$, such that $\mathbf{Q}\mathbf{v} = \mathbf{v} - \mathbf{P}\mathbf{v} = \mathbf{v} - \mathbf{v}_1 = \mathbf{v}_2 = \Pi(\mathbf{v}|V_2)$. By direct calculation:

$$\mathbf{P}\mathbf{Q} = \mathbf{P}(\mathbf{I} - \mathbf{P}) = \mathbf{P} - \mathbf{P}^2 = \mathbf{P} - \mathbf{P} = \mathbf{0}.$$

Likewise, $\mathbf{Q}\mathbf{P} = (\mathbf{I} - \mathbf{P})\mathbf{P} = \mathbf{P} - \mathbf{P}^2 = \mathbf{P} - \mathbf{P} = \mathbf{0}$. Thus:

$$\Pi\{\Pi(\mathbf{v}|V_1)|V_2\} = \mathbf{0} = \Pi\{\Pi(\mathbf{v}|V_2)|V_1\}$$

when V_1 is linearly independent of V_2 , or equivalently when $V_1 \cap V_2 = \{\mathbf{0}\}$. ■

Proposition 3.4. Suppose $\mathbf{v} \in V$ and $\mathbf{P}^2 = \mathbf{P}$, then $V = V_1 \oplus V_2$ where $V_1 = \text{im}(\mathbf{P})$ and $V_2 = \text{im}(\mathbf{I} - \mathbf{P})$. $\blacklozenge$

Proof. Any $\mathbf{v}$ in V is expressible as:

$$\mathbf{v} = (\mathbf{I} - \mathbf{P} + \mathbf{P})\mathbf{v} = (\mathbf{I} - \mathbf{P})\mathbf{v} + \mathbf{P}\mathbf{v}.$$

Thus $V = V_2 + V_1$. To verify that V is the direct sum, it suffices to show $V_1 \cap V_2 = \{\mathbf{0}\}$. Suppose $\mathbf{u} \in V_1 \cap V_2$. Since $\mathbf{u} \in V_2$, there exists a coefficient vector $\boldsymbol{\alpha}$ such that $\mathbf{u} = (\mathbf{I} - \mathbf{P})\boldsymbol{\alpha}$. Moreover, since $\mathbf{u} \in V_1$, $\mathbf{u} = \mathbf{P}\mathbf{u}$. Now:

$$\mathbf{u} = \mathbf{P}\mathbf{u} = \mathbf{P}(\mathbf{I} - \mathbf{P})\boldsymbol{\alpha} = (\mathbf{P} - \mathbf{P}^2)\boldsymbol{\alpha} = (\mathbf{P} - \mathbf{P})\boldsymbol{\alpha} = \mathbf{0}.$$

■

Proposition 3.5. All eigenvalues of a projection matrix are either 0 or 1. $\blacklozenge$

Proof. Suppose $\mathbf{P}$ is a projection matrix with eigenvector $\mathbf{u}$, then $\mathbf{P}\mathbf{u} = \lambda\mathbf{u}$ for some eigenvalue λ . Left multiplying by $\mathbf{P}$:

$$\mathbf{P}^2\mathbf{u} = \mathbf{P}\lambda\mathbf{u} = \lambda\mathbf{P}\mathbf{u} = \lambda^2\mathbf{u}.$$

Yet by idempotency of $\mathbf{P}$, $\mathbf{P}^2\mathbf{u} = \mathbf{P}\mathbf{u} = \lambda\mathbf{u}$. Thus $\lambda\mathbf{u} = \lambda^2\mathbf{u}$, so $\lambda \in \{0, 1\}$. $\blacksquare$

Hilbert Space

Definition 4.1 (Hilbert Space). A Hilbert space $\mathcal{H}$ is a *complete linear space* with an *inner product norm*:

$$\|\mathbf{v}\|^2 = \langle \mathbf{v}, \mathbf{v} \rangle, \text{ for } \mathbf{v} \in \mathcal{H}.$$

Euclidean space is a finite dimensional Hilbert space. $\blacksquare$

4.1 Orthogonal Complement

Definition 4.2. The **orthogonal complement** $V^\perp$ to a subspace V consists of those vectors orthogonal to each element of V :

$$V^\perp = \{\mathbf{u} \in \Omega : \langle \mathbf{u}, \mathbf{v} \rangle = 0 \text{ for } \forall \mathbf{v} \in V\}.$$

■

Proposition 4.1. The orthogonal complement $V^\perp$ is (always) a closed subspace. $\blacklozenge$

Proof. Suppose $\mathbf{v} \in V$. If $\mathbf{u}_1$ and $\mathbf{u}_2$ are two vectors in $V^\perp$, then any linear combination $\mathbf{u} = \alpha_1 \mathbf{u}_1 + \alpha_2 \mathbf{u}_2$ remains in $V^\perp$ since:

$$\langle \mathbf{u}, \mathbf{v} \rangle = \alpha_1 \langle \mathbf{u}_1, \mathbf{v} \rangle + \alpha_2 \langle \mathbf{u}_2, \mathbf{v} \rangle = 0.$$

Suppose $(\mathbf{u}_n)$ is a Cauchy sequence in $V^\perp$, then $(\mathbf{u}_n)$ converges to some limit $\mathbf{u}$. By continuity of the inner product:

$$\langle \mathbf{u}, \mathbf{v} \rangle = \lim_{n \rightarrow \infty} \langle \mathbf{u}_n, \mathbf{v} \rangle = 0.$$

■

Proposition 4.2. The following inclusions hold:

i. $V \subseteq V^{\perp\perp}$.

ii. If $U \subseteq V$, then $V^\perp \subseteq U^\perp$.

◆

Proof. (i.) If $\mathbf{v} \in V$, then $\langle \mathbf{v}, \mathbf{u} \rangle = 0$ for $\forall \mathbf{u} \in V^\perp$, therefore $\mathbf{v} \in V^{\perp\perp}$.

(ii.) If $\mathbf{v}_\perp \in V^\perp$, then $\mathbf{v}_\perp$ is orthogonal to each $\mathbf{v}$ in V . Now $U \subseteq V$, so $\mathbf{v}_\perp$ is orthogonal to each $\mathbf{u} \in U$. Conclude $\mathbf{v}_\perp \in U^\perp$. ■

4.2 Finite Closure

Theorem 4.1. Suppose V is a *finite* subspace of a Hilbert space $\mathcal{H}$, then V is complete, and therefore closed. □

Proof. Suppose V is a finite dimensional subspace with basis $\mathbf{B} = (\mathbf{b}_1, \dots, \mathbf{b}_n)$, then every $\mathbf{v} \in V$ is expressible as $\mathbf{v} = \alpha_1 \mathbf{b}_1 + \dots + \alpha_n \mathbf{b}_n$ for some coefficients $(\alpha_1, \dots, \alpha_n)$. There exists a bijection $T : V \rightarrow \mathbb{F}^n$ such that $T(\mathbf{v}) = (\alpha_1, \dots, \alpha_n)$, where $\mathbb{F}$ is a complete field. Moreover, this mapping is linear since:

$$T(\mathbf{v}_1 + \mathbf{v}_2) = (\alpha_1 + \beta_1, \dots, \alpha_n + \beta_n) = (\alpha_1, \dots, \alpha_n) + (\beta_1, \dots, \beta_n) = T(\mathbf{v}_1) + T(\mathbf{v}_2).$$

Suppose $(\mathbf{v}_n)$ is a Cauchy sequence in V . For $\epsilon > 0$, define $\delta = \epsilon / \|T\|$, where:

$$\|T\| = \sup_{\{\mathbf{v} \neq \mathbf{0}\}} \frac{\|T(\mathbf{x})\|}{\|\mathbf{x}\|}.$$

Since $\mathbf{v}_n$ is Cauchy, there exist a threshold ν such that when $n, m \geq \nu$:

$$\|\mathbf{v}_n - \mathbf{v}_m\| < \frac{\epsilon}{\|T\|}.$$

Now for $n, m \geq \nu$:

$$\|T(\mathbf{v}_n) - T(\mathbf{v}_m)\| = \|T(\mathbf{v}_n - \mathbf{v}_m)\| \leq \|T\| \cdot \|\mathbf{v}_n - \mathbf{v}_m\| = \epsilon.$$

Conclude that $\{T_n \equiv T(\mathbf{v}_n)\}$ is Cauchy. Since $\mathbb{F}^n$ is complete, (T_n) converges to some limit $T_0 \in \mathbb{F}^n$, and:

$$\lim_{n \rightarrow \infty} \|T(\mathbf{v}_n) - T_0\| = 0.$$

Since T is a bijection, there exists $\mathbf{v}_0 \in V$ such that $T(\mathbf{v}_0) = T_0$. Now:

$$0 \leq \lim_{n \rightarrow \infty} \|\mathbf{v}_n - \mathbf{v}_0\| = \lim_{n \rightarrow \infty} \|T^{-1}T(\mathbf{v}_n) - T^{-1}T_0\| \leq \|T^{-1}\| \lim_{n \rightarrow \infty} \|T(\mathbf{v}_n) - T_0\| = 0.$$

Since $(\mathbf{v}_n)$ converges to a limit in V , conclude that V is complete. ■

Orthogonal Projection

Definition 5.1. Let V denote a closed subspace of a Hilbert space $\mathcal{H}$. The **orthogonal projection** of $\mathbf{u} \in \mathcal{H}$ onto V is the element $\Pi_0(\mathbf{u}|V)$ with these properties:

- i. the projection resides in V : $\Pi_0(\mathbf{u}|V) \in V$.
 - ii. the residual is orthogonal to V : $\mathbf{u} - \Pi_0(\mathbf{u}|V) \in V^\perp$.
-

Remark 5.1. The notation Π_0 distinguishes the orthogonal projection from the potentially oblique projection Π . ◆

Lemma 5.1.

$$\|\mathbf{x} + \mathbf{y}\|^2 + \|\mathbf{x} - \mathbf{y}\|^2 = 2\|\mathbf{x}\|^2 + 2\|\mathbf{y}\|^2. \quad (5.1)$$

Proof is by direct expansion. ■

Theorem 5.1 (Projection Theorem). Suppose V is a closed subspace of a Hilbert space $\mathcal{H}$. For any $\mathbf{u} \in \mathcal{H}$, there *exists* a *unique* closest element in V to $\mathbf{u}$:

$$\hat{\mathbf{u}} = \arg \min_{\mathbf{v} \in V} \|\mathbf{v} - \mathbf{u}\|^2,$$

and $\hat{\mathbf{u}}$ is the *orthogonal projection* of $\mathbf{u}$ onto V . □

Proof. (EXISTENCE) If $\mathbf{u} \in V$, then $\hat{\mathbf{u}} = \mathbf{u}$. Suppose not. Let $\delta^2 = \inf_{\mathbf{v} \in V} \|\mathbf{v} - \mathbf{u}\|^2$. Construct a sequence $\mathbf{v}_n \in V$ such that:

$$\|\mathbf{v}_n - \mathbf{u}\|^2 \leq \delta^2 + \frac{1}{n}.$$

By (5.1):

$$\|(\mathbf{v}_n - \mathbf{u}) + (\mathbf{u} - \mathbf{v}_m)\|^2 + \|(\mathbf{v}_n - \mathbf{u}) - (\mathbf{u} - \mathbf{v}_m)\|^2 = 2\|\mathbf{v}_n - \mathbf{u}\|^2 + 2\|\mathbf{u} - \mathbf{v}_m\|^2.$$

Upon rearranging:

$$\begin{aligned} \|\mathbf{v}_n - \mathbf{v}_m\|^2 + 4\left\|\frac{\mathbf{v}_n + \mathbf{v}_m}{2} - \mathbf{u}\right\|^2 &= 2\|\mathbf{v}_n - \mathbf{u}\|^2 + 2\|\mathbf{u} - \mathbf{v}_m\|^2, \\ \|\mathbf{v}_n - \mathbf{v}_m\|^2 &= 2\|\mathbf{v}_n - \mathbf{u}\|^2 + 2\|\mathbf{u} - \mathbf{v}_m\|^2 - 4\left\|\frac{\mathbf{v}_n + \mathbf{v}_m}{2} - \mathbf{u}\right\|^2. \end{aligned}$$

Now, for all $(n, m) \in \mathbb{N}^2$, the midpoint $\boldsymbol{\mu}_{nm} = (\mathbf{v}_n + \mathbf{v}_m)/2$ is in V since V is convex. Therefore, $\|\boldsymbol{\mu}_{nm} - \mathbf{u}\|^2 \geq \delta^2$, and:

$$\|\mathbf{v}_n - \mathbf{v}_m\|^2 \leq 2\|\mathbf{v}_n - \mathbf{u}\|^2 + 2\|\mathbf{v}_m - \mathbf{u}\|^2 - 4\delta^2.$$

By construction $\lim_{n \rightarrow \infty} \|\mathbf{v}_n - \mathbf{u}\|^2 = \delta^2$, thus:

$$\lim_{(n,m) \rightarrow \infty} \|\mathbf{v}_n - \mathbf{v}_m\|^2 \leq 2\delta^2 + 2\delta^2 - 4\delta^2 = 0.$$

Conclude that $(\mathbf{v}_n)$ is a Cauchy sequence. Since V is a closed subspace of a Hilbert space, $(\mathbf{v}_n)$ converges to a limit in V . Call this limit $\hat{\mathbf{u}}$. By continuity of the norm:

$$\|\hat{\mathbf{u}} - \mathbf{u}\|^2 = \lim_{n \rightarrow \infty} \|\mathbf{v}_n - \mathbf{u}\|^2 \leq \lim_{n \rightarrow \infty} \left(\delta^2 + \frac{1}{n}\right) = \delta^2.$$

Moreover since $\hat{\mathbf{u}} \in V$, $\|\hat{\mathbf{u}} - \mathbf{u}\|^2 \geq \delta^2$ by definition of the infimum. Conclude that $\|\hat{\mathbf{u}} - \mathbf{u}\|^2 = \delta^2$, and therefore $\hat{\mathbf{u}} = \arg \min_{\mathbf{v} \in V} \|\mathbf{v} - \mathbf{u}\|^2$.

(ORTHOGONAL PROJECTION) Since $\hat{\mathbf{u}}$ is in V , to establish $\hat{\mathbf{u}}$ is the orthogonal projection of $\mathbf{u}$ onto V , it remains to show that $\mathbf{u} - \hat{\mathbf{u}}$ is in $V^\perp$. Suppose to the contrary that there exists $\mathbf{v}$ in V not orthogonal to $\mathbf{u} - \hat{\mathbf{u}}$. WLOG let $\|\mathbf{v}\|^2 = 1$, and set $\theta = \langle \mathbf{u} - \hat{\mathbf{u}}, \mathbf{v} \rangle$. Define $\tilde{\mathbf{u}} = \theta \mathbf{v} + \hat{\mathbf{u}}$, which is in V . Then:

$$\begin{aligned} \|\tilde{\mathbf{u}} - \mathbf{u}\|^2 &= \|\theta \mathbf{v} + \hat{\mathbf{u}} - \mathbf{u}\|^2 = \|\theta \mathbf{v}\|^2 + \|\hat{\mathbf{u}} - \mathbf{u}\|^2 + 2\theta \langle \mathbf{v}, \hat{\mathbf{u}} - \mathbf{u} \rangle \\ &= \theta^2 + \|\hat{\mathbf{u}} - \mathbf{u}\|^2 - 2\theta^2 = \|\hat{\mathbf{u}} - \mathbf{u}\|^2 - \theta^2 \leq \|\hat{\mathbf{u}} - \mathbf{u}\|^2. \end{aligned}$$

Thus, if $\mathbf{u} - \hat{\mathbf{u}}$ is not in $V^\perp$, then $\hat{\mathbf{u}}$ is not the closest point in V to $\mathbf{u}$. Since $\hat{\mathbf{u}} \in V$ and $\mathbf{u} - \hat{\mathbf{u}} \in V^\perp$ if $\hat{\mathbf{u}} = \arg \min_{\mathbf{v} \in V} \|\mathbf{v} - \mathbf{u}\|^2$, conclude that $\hat{\mathbf{u}} = \Pi_0(\mathbf{u}|V)$.

(UNIQUENESS) Suppose $\hat{\mathbf{u}}_1$ and $\hat{\mathbf{u}}_2$ are both orthogonal projections of $\mathbf{u}$ onto V , then $\mathbf{u} - \hat{\mathbf{u}}_1 \in V^\perp$ and $\mathbf{u} - \hat{\mathbf{u}}_2 \in V^\perp$. Thus $(\mathbf{u} - \hat{\mathbf{u}}_1) - (\mathbf{u} - \hat{\mathbf{u}}_2) = \hat{\mathbf{u}}_2 - \hat{\mathbf{u}}_1 \in V^\perp$. But $\hat{\mathbf{u}}_2 \in V$ and $\hat{\mathbf{u}}_1 \in V$, so $\hat{\mathbf{u}}_2 - \hat{\mathbf{u}}_1 \in V \cap V^\perp = \{\mathbf{0}\}$. Conclude $\hat{\mathbf{u}}_1 = \hat{\mathbf{u}}_2$. ■

Corollary 5.1. If V is a closed linear subspace of a Hilbert space $\mathcal{H}$, then:

$$\mathcal{H} = V \oplus V^\perp.$$



Proposition 5.1. Suppose V is a closed subspace of Hilbert space $\mathcal{H}$ and that V is the direct sum of two orthogonal subspaces: $V = V_0 \oplus V_1$ with $V_0 \perp V_1$. Then, for any $\mathbf{u} \in \mathcal{H}$, the projection onto V is the sum of the projections onto V_0 and onto V_1 :

$$\Pi_0(\mathbf{u}|V) = \Pi_0(\mathbf{u}|V_0) + \Pi_0(\mathbf{u}|V_1).$$



Proof. Let $\hat{\mathbf{u}}_0$ and $\hat{\mathbf{u}}_1$ denote orthogonal projection onto V_0 and V_1 . Consider the sum $\hat{\mathbf{u}} = \hat{\mathbf{u}}_0 + \hat{\mathbf{u}}_1$. Since $V = V_0 \oplus V_1$, $\hat{\mathbf{u}} \in V$. Moreover, for any $\mathbf{v} = \mathbf{v}_0 + \mathbf{v}_1 \in V$:

$$\begin{aligned} \langle \mathbf{u} - \hat{\mathbf{u}}, \mathbf{v} \rangle &= \langle \mathbf{u} - \hat{\mathbf{u}}_0 - \hat{\mathbf{u}}_1, \mathbf{v}_0 + \mathbf{v}_1 \rangle \\ &= \langle \mathbf{u} - \hat{\mathbf{u}}_0, \mathbf{v}_0 \rangle - \langle \hat{\mathbf{u}}_1, \mathbf{v}_0 \rangle + \langle \mathbf{u} - \hat{\mathbf{u}}_1, \mathbf{v}_1 \rangle - \langle \hat{\mathbf{u}}_0, \mathbf{v}_1 \rangle = 0. \end{aligned}$$

Since $\hat{\mathbf{u}} \in V$ and $\mathbf{u} - \hat{\mathbf{u}} \in V^\perp$, $\hat{\mathbf{u}}$ is the orthogonal projection onto V . ■

Proposition 5.2. Suppose $\mathbf{P}$ is the matrix for orthogonal projection onto a closed linear subspace V . $\mathbf{P}$ is **self-adjoint**:

$$\langle \mathbf{P}\mathbf{u}_1, \mathbf{u}_2 \rangle = \langle \mathbf{P}\mathbf{u}_1, \mathbf{P}\mathbf{u}_2 \rangle = \langle \mathbf{u}_1, \mathbf{P}\mathbf{u}_2 \rangle.$$



Proof. Since $\mathbf{u}_2 = \mathbf{P}\mathbf{u}_2 + (\mathbf{I} - \mathbf{P})\mathbf{u}_2$ with $\mathbf{P}\mathbf{u}_2 = \hat{\mathbf{u}}_2 \in V$ and $(\mathbf{I} - \mathbf{P})\mathbf{u}_2 = \mathbf{u}_2 - \hat{\mathbf{u}}_2 \in V^\perp$,

$$\langle \mathbf{P}\mathbf{u}_1, \mathbf{u}_2 \rangle = \langle \mathbf{P}\mathbf{u}_1, \mathbf{P}\mathbf{u}_2 + (\mathbf{I} - \mathbf{P})\mathbf{u}_2 \rangle = \langle \mathbf{P}\mathbf{u}_1, \mathbf{P}\mathbf{u}_2 \rangle.$$

Writing $\mathbf{u}_1 = \mathbf{P}\mathbf{u}_1 + (\mathbf{I} - \mathbf{P})\mathbf{u}_1$ likewise gives:

$$\langle \mathbf{u}_1, \mathbf{P}\mathbf{u}_2 \rangle = \langle \mathbf{P}\mathbf{u}_1 + (\mathbf{I} - \mathbf{P})\mathbf{u}_1, \mathbf{P}\mathbf{u}_2 \rangle = \langle \mathbf{P}\mathbf{u}_1, \mathbf{P}\mathbf{u}_2 \rangle.$$



Corollary 5.2. For Euclidean space:

$$\mathbf{u}'_1 \mathbf{P}' \mathbf{u}_2 = \langle \mathbf{P}\mathbf{u}_1, \mathbf{u}_2 \rangle = \langle \mathbf{u}_1, \mathbf{P}\mathbf{u}_2 \rangle = \mathbf{u}'_1 \mathbf{P} \mathbf{u}_2.$$

That is, $\mathbf{P}$ is symmetric. ♣

5.1 Euclidean Space

Proposition 5.3. Suppose $V_0 \subseteq V$ are closed linear subspaces of Euclidean space. Let $\mathbf{P}_0$ and $\mathbf{P}$ denote orthogonal projection onto V_0 and V , then:

$$\mathbf{P}\mathbf{P}_0 = \mathbf{P}_0 = \mathbf{P}_0\mathbf{P}.$$

That is, projection onto V_0 is equivalent to projection onto V , followed by projection onto $V_0 \subseteq V$. The operator $\mathbf{P}_0\mathbf{P}$ is an *iterated projection*. $\blacklozenge$

Proof. For any $\mathbf{v} \in \mathcal{H}$, since $\mathbf{P}_0\mathbf{v} \in V_0 \subseteq V$, $\mathbf{P}(\mathbf{P}_0\mathbf{v}) = \mathbf{P}_0\mathbf{v}$. Thus, $\mathbf{P}\mathbf{P}_0 = \mathbf{P}_0$. Further, since $\mathbf{P}_0$ and $\mathbf{P}$ are symmetric in Euclidean space:

$$\mathbf{P}_0 = \mathbf{P}_0' = (\mathbf{P}\mathbf{P}_0)' = \mathbf{P}_0'\mathbf{P}' = \mathbf{P}_0\mathbf{P}.$$

■

Proposition 5.4. Suppose $V_0 \subseteq V$ are closed linear subspaces of Euclidean space. Let $\mathbf{P}_0$ and $\mathbf{P}$ denote orthogonal projection onto V_0 and V . Define V_1 as the subspace of V that is orthogonal to V_0 , $V_1 = V_0^\perp \cap V$. Then:

- i. $V = V_0 \oplus V_1$ with $V_0 \perp V_1$.
- ii. $\mathbf{P}_1 = \mathbf{P} - \mathbf{P}_0$, where $\mathbf{P}_1$ is orthogonal projection onto V_1 .

◆

Proof. (i.) Suppose $\mathbf{v} \in V$, and let $\hat{\mathbf{v}}_0 = \mathbf{P}_0\mathbf{v}$. Since $\mathbf{v} - \hat{\mathbf{v}}_0 \in V$ and $\mathbf{v} - \hat{\mathbf{v}}_0 \in V_0^\perp$, $\mathbf{v} - \hat{\mathbf{v}}_0 \in V_1$. Writing $\mathbf{v} = \hat{\mathbf{v}}_0 + (\mathbf{v} - \hat{\mathbf{v}}_0)$ demonstrates $\mathbf{v} \in V_0 + V_1$. Conversely, since $V_0 \subseteq V$ and $V_1 \subseteq V$, $V_0 + V_1 \subseteq V$. Moreover, since $V_1 = V_0^\perp \cap V \subseteq V_0^\perp$, $V_1 \perp V_0$, and the sum $V = V_0 \oplus V_1$ is direct.

(ii.) Since V is the direct sum of two orthogonal subspaces, the orthogonal projection onto V is expressible as $\mathbf{P} = \mathbf{P}_0 + \mathbf{P}_1$. ■

Example 5.1. Suppose $\mathbb{E}^n$ is Euclidean space and $V \subset \mathbb{E}^n$ is a linear subspace, then $\mathbb{E}^n$ is expressible as $V \oplus V^\perp$. Every $\mathbf{u} \in \mathbb{E}^n$ is uniquely expressible as $\hat{\mathbf{u}} + \mathbf{u}_\perp$, where $\hat{\mathbf{u}} = \mathbf{P}\mathbf{u}$ is the orthogonal projection onto V , and $\mathbf{u}_\perp = (\mathbf{I} - \mathbf{P})\mathbf{u}$ is the orthogonal projection onto $V^\perp$. The orthogonal projections are guaranteed to exist by closure of V and $V^\perp$. Moreover, the two projections are orthogonal $\langle \hat{\mathbf{u}}, \mathbf{u}_\perp \rangle = 0$. ♠